BRIDGES IN MATHEMATICS • GRADE 2 • UNIT 2

Place Value & Bundling Tens into Hundreds

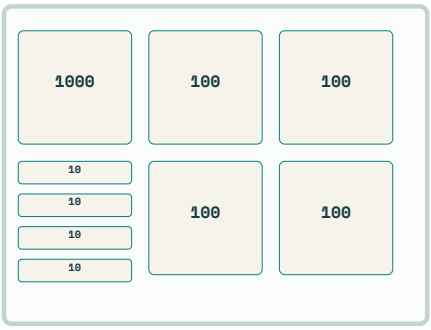
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.2.NBT.A.1

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Place Value Base Ten

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Group 10 unit cubes into 1 rod. Group 10 rods into 1 flat (100).

Sketch Model on Paper:
Draw squares for hundreds flats, lines for tens rods, dots for ones.

3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

1.	10	X: 10	10 units per rod int
2.	ENTER	Y: 10 | X: 10	Stack lift
3.	10	Y: 10 | X: 10	10 rods per flat int
4.	×	X: 100	Multiply: 10 × 10 =

EXPECTED LCD X: 100 (1 hundred flat)

1 **Combine 4 hundreds (400) and 5 tens (50).**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

2 **Combine 6 hundreds, 3 tens, and 8 ones.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

3 **Subtract 1 ten from 300.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

4 **What is 100 more than 460?**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-6

Station Diagnostic & Solution Key: Unit 2

Place Value & Bundling Tens into Hundreds • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Place Value Base-Ten Studio (Blocks & Flats)

CAD Blueprint: place_value_base_ten

Target Device: SC-6 Elementary

CAD / STL Source File:

Education/scad/lib/place_value_base_ten.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

100 Yellow Flats & 10 Green Rods (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Combine 4 hundreds (400) and 5 tens (50).** Standard: CCSS.MATH.2.NBT.A.1	450	[4][0][0] [ENTER] [5][0] [+]
2	**Combine 6 hundreds, 3 tens, and 8 ones.** Standard: CCSS.MATH.2.NBT.A.1	638	[6][0][0] [ENTER] [3][0] [+] [8] [+]
3	**Subtract 1 ten from 300.** Standard: CCSS.MATH.2.NBT.A.1	290	[3][0][0] [ENTER] [1][0] [-]
4	**What is 100 more than 460?** Standard: CCSS.MATH.2.NBT.A.1	560	[4][6][0] [ENTER] [1][0][0] [+]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for confusing 100 with 10 tens.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.